# Laurence Martin

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## Profile

Test Automation Engineer with 7 years of software engineering experience spanning industrial control systems and QA automation. Design end-to-end test automation architecture for marine electronics, from build ingestion and CI/CD pipelines to test execution, reporting and defect resolution, bridging development, firmware and QA teams. Strong in Python, PyTest, Selenium/Playwright, Windows, Docker and Linux, with an embedded C/C++ background and a First Class MEng in Electronic Engineering.

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## Technical Skills

**Languages:** Python, C, C++, C#, Kotlin, JavaScript, HTML/CSS, SQL, Shell

**Testing:** PyTest, Selenium, Playwright, Allure, ReportPortal, regression, soak / stability, integration and E2E testing

**DevOps:** CI/CD pipelines (GitLab CI, GitHub Actions), Docker, Git, Linux / Windows

**Platforms:** Django, PostgreSQL, Redis, Celery, SQLite, Android (Kotlin, Jetpack Compose, MVVM, Room + SQLCipher, OkHttp, Chaquopy), Gemini API

**Tools:** Jira, Confluence, SSH, Visual Studio, VS Code, AI-assisted development

**Industrial:** TCP/IP, Modbus TCP / Serial, Moxa, Siemens PLC (PN/PN coupler), network diagnostics

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## Professional Experience

### Test Automation Engineer – Automation Architecture & CI/CD — 2024 - Present

*Alten (Consultant to Raymarine)*

- Architect and own the end-to-end test automation system for Raymarine thermal camera and marine electronics products, from build release through test execution, reporting and defect resolution
- Design and integrate CI/CD pipelines across Windows and Linux environments, connecting development, firmware and QA teams so each new build is automatically deployed, tested and reported
- Integrate automated test execution with Jira test management, giving stakeholders live release-readiness visibility and reducing manual reporting by 90%
- Build soak, regression and integration test suites with PyTest and ReportPortal / Allure for in-development and released products, providing scheduled build-quality feedback
- Develop real-user E2E UI automation (Selenium, Playwright) for web and app interfaces across multiple product lines
- Built and maintain a Django test orchestration server (PostgreSQL, Redis, Docker) to control devices, monitor network traffic and capture diagnostics via SSH and network statistics
- Act as the technical link across departments on test strategy, tooling, standards and product trials

**Software Engineer - SCADA / Controls (Field Service)** **2019 - 2024**

*EPROD ESCADA*

- Developed and commissioned SCADA process and machine control systems for the corrugated cardboard industry, delivering 20+ projects across 15 countries
- Designed project-specific desktop control applications in C/C++/C# from technical contracts and electrical drawings, collaborating with project managers and electrical / mechanical engineers
- Engineered device interfaces over Modbus TCP and serial (Moxa) and PLC integration via Siemens PN/PN couplers
- Managed customer communication, on-site installation and international travel logistics for myself and colleagues

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## Education

**MEng Electronic Engineering with Music Technology Systems** **2015 - 2019**

*University of York – First Class Honours*

- Final-year project (1st – 73%): applied evolutionary algorithms to analyse speech deficits in Parkinson's Disease patients before and after Deep Brain Stimulation (DBS) surgery

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## Projects

### Momex

- Android personal finance app combining multiple banks and investment platforms into one overview via the Endute and Trading212 APIs
- AI-powered database reports and plain-English search over financial data using the Google Gemini API
- Encrypted local storage and backups (SQLCipher AES-256), biometric login and weekly spending reminders via WorkManager
- Balance and spending trend charts with transaction filtering, built with Vico

### Auto Stock Trader

- Django platform with a PyQt5 controller to view, analyse and execute trades on US markets via the Interactive Brokers (IBKR) API
- Live ranking of stocks with entry / exit recommendations and an optional fully automated trading mode
- Machine learning analysis of historic trades to identify the factors driving success and loss rates
- Asynchronous request and callback handling for the IBKR API using Celery, Django Channels and Redis

### Password Manager

- Android password manager for websites, apps and systems, able to launch third-party apps from one central hub
- Fully encrypted JSON export / import for backup and restore, with biometric login secured via Android security libraries

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## Activities and Interests

- Multi-instrumentalist (guitar, bass, piano, drums), played in multiple bands and compose across metal, rock, chiptune and EDM
- Stay active with the gym several times a week and 5-a-side football
- Enjoy travelling, exploring new places and food, and gaming across genres and platforms